

A class-built view of Smart Logistics

Lecture 01, taught by Professor Wai Kin (Victor) Chan, introduced the history of supply chains and AI, examined AI readiness, and used fresh-food distribution, Flexport and S.F. Holding as case studies. The class then connected those ideas into a working logistics decision system.

30

unique submissions

11

named example geographies

17

mentioned both AI and optimization in Q3

Class-derived definition. Smart Logistics is a connected, data-informed operating system that senses conditions, shares usable information, predicts possible futures, optimizes decisions under constraints, acts through people and technology, and learns from outcomes. Skills, governance and resilience shape every stage.

The technology stack students named

The themes that appeared most often in the class's keywords.

AI		24/30
Optimization		22/30
IoT		20/30
Automation		19/30
Digital Twin		17/30
Big Data		15/30
Real-time Data		7/30
Connectivity		7/30
LLM		5/30
Prediction		5/30

Why the loop matters

Sense and connect: IoT and real-time information matter when they can move across firms and functions. A sensor reading in a silo is weak evidence for action.

Predict and optimize: AI estimates demand, delay or risk; operations research and optimization choose feasible routes, schedules and resources. The distinction appears explicitly in student insights.

Act and learn: Automation executes some choices; people remain responsible for exceptions. Digital twins and feedback help test and revise decisions.

Two co-mentions reinforce the system view: **17 of 30** named AI and optimization together; **15 of 30** named IoT and digital twins together. These combinations reveal how the class connected sensing and decisions.

A more adaptive future, with people in control

Across the 5–10 year outlooks, students anticipated more predictive and adaptive planning, physical automation, stronger shared data infrastructure and wider use of digital twins and AI agents. They also raised workforce skills, privacy, reliability, accountability and regulation. Some treated sustainability and resilience as objectives.

Examples across regions

A view across delivery networks, ports, warehousing and policy.

EXAMPLE GEOGRAPHY	STUDENT-CITED CASE	TOPIC DISCUSSED
Vietnam	Viettel Post sorting	Warehouse automation
China	JD Logistics delivery	Autonomous delivery
Indonesia	Cikarang 5G warehouse	Connectivity + AGVs
Morocco	Tanger Med assistants	Port + AI services
Chile	San Antonio port	Terminal + rail links
Brazil	APM Suape terminal	Port electrification
Russia	Yandex delivery robots	Last-mile robotics
Austria	Post Vienna-Inzersdorf	Sorting operations
Bangladesh	National Logistics Policy	Policy / infrastructure

Distinctive class insights

"Start with data, not tools."	"AI predicts; Operations Research decides."
"Automate the plan, but keep a human on the button."	"Prediction creates value only when joined to optimization and operational execution."

Teaching signal

Q2 feedback asked for practical exercises, technical explanation, deeper cases, implementation constraints and discussion. Learners want to see how a forecast becomes a workable logistics decision.

What to carry forward

The strongest lesson is operational: a forecast only becomes useful when it informs a feasible choice and reaches execution. Future lectures on transportation and warehousing can return to the same chain of observation, decision and feedback, while testing where human judgement belongs.